

7. Recent Progress and Updated Project Direction

7.1 MFCC Extraction Alignment and Validation Update

During continued development, additional inconsistencies were identified between MFCC features extracted in Python and those computed on the ESP32. To address this, MFCC extraction was implemented directly on the ESP32, allowing direct comparison between embedded C++ MFCCs and the Python training pipeline.

A configuration flag was added to the Python code to allow MFCC extraction using Librosa. This made it possible to compare Python-based MFCCs with those generated on the ESP32 and verify consistency between training-time and deployment-time feature distributions. As a result, alignment between the two pipelines was improved, increasing confidence in real-time embedded performance.

7.2 Dataset Labeling Issue and Accuracy Improvement

When running validation on a dataset larger than the one used in the initial report, a significant drop in accuracy was observed. Further investigation showed that this issue was caused by incorrect label handling rather than model behavior.

Specifically, some audio files were incorrectly classified as “other” instead of “anger” due to case sensitivity in Python file handling. This labeling mismatch caused a large number of samples to be assigned to the wrong class, significantly affecting the reported accuracy.

After correcting and reordering the mislabeled files, validation was repeated. The updated results showed a much higher and more reliable accuracy, confirming that the previous performance drop was due to a dataset issue and that the current ML model performs better than initially reported.

7.3 Updated Project Direction: IoT-Based System

Based on the current state of the system, the project will continue as an IoT-oriented embedded system. The anger detection model is treated as a sensing component that operates in real time on the ESP32 and can be integrated into a larger interactive system.

The focus is no longer only on classification accuracy, but also on system behavior, user interaction, and feedback.

7.4 Planned Features and System Extensions

The following features are currently planned or under development:

- **Detection Cancellation Button**

A physical button will be added to allow the user to cancel a detection event after it occurs. If the user cancels the detection, no notification or external action will be triggered.

- **Pulse Sensor Integration**

A pulse sensor will be integrated alongside the audio-based model to provide an additional physiological signal. This is intended to improve robustness of anger detection by combining audio features with heart-rate information.

- **Mobile Application and Cloud Connectivity**

The ESP32 will be connected to a mobile application using Firebase. This will enable:

- Sending detection events from the ESP32 to the application
- Logging events for monitoring and analysis
- Providing feedback from the application to the device

Depending on hardware limitations, feedback from the application may be implemented as either text messages or simple audio cues.

7.5 Current Status Summary

- ML model accuracy has been improved after correcting dataset labeling issues
- MFCC consistency between Python and ESP32 implementations has been strengthened
- The project direction is now clearly IoT-focused
- Ongoing work emphasizes interaction, feedback, and multi-sensor support